

Bench Meter Ladder

The complete ladder program for the bench meter: scaling, overload and fault flags, 100 ms averaging, mode select, HOLD, REL (zero), MIN/MAX and the continuity beeper, plus the analog module setup and the C-more tags that go with it. Enter it rung by rung in the CLICK software.

1 · Analog module setup

In CLICK software, open **Setup** → **System Configuration**, select the C0-4AD2DA-2 (slot 1), and open its setup:

- **Inputs CH1–CH4:** enabled, 0–10 V, scaled to engineering units **0.000** → **10.000**, stored in DF1–DF4. Every value in the ladder is then in volts at the terminal.
- **Output CH1:** from DF5, scaled **0.000** → **10.000** = 0–10 V. **Output CH2:** disabled.
- If the software assigns different DF addresses, keep its choice and swap those addresses into the ladder.
- **The C0-02DD2-D's own analog (AD1/AD2, DA1/DA2) is not used.** It's only 0–5 V or 4–20 mA at 12-bit, so all measuring stays on the C0-4AD2DA-2. Check *Setup* → *CPU Built-in I/O Setup* to see which registers the built-in channels took, so you don't mix them up with the module's DF1–DF5.
- **Setup** → **Com Port Setup:** Port 2 = Modbus RTU, node 1, 38400, 8-O-1 for the panel. Leave Port 1 at its defaults for programming.
- **Setup** → **Memory Setup:** make sure DF20 (the DC calibration factor) is in the retentive range, so it survives a power cycle.

2 · Address & nickname map

Type these nicknames into the Address Picker first. Then the rungs below read the same way on your screen.

ADDRESS	NICKNAME	WHAT IT IS
PHYSICAL I/O		
Y001	BEEPER	Continuity buzzer ECX2070-24 · sourcing output, 0.1 A max
ANALOG (FLOAT, VOLTS)		

ADDRESS	NICKNAME	WHAT IT IS
DF1	AI_ACV	CH1: VACT loop across R1, 1–5 V
DF2	AI_DCV	CH2: DC divider, 0–10 V
DF3	AI_OHM	CH3: volts across Rx
DF4	AI_EXCITE	CH4: actual excitation, about 10 V
DF5	AO_EXCITE	OUT CH1 command, 10.0
CALCULATED (FLOAT)		
DF10 / 11 / 12	ACV_RAW · DCV_RAW · OHM_RAW	Scaled readings every scan
DF13 / 14 / 15	ACV_AVG · DCV_AVG · OHM_AVG	Filtered readings, updated every 100 ms
DF16	OHM_OPEN_THR	99% of excitation (the open-leads threshold)
DF20	DCV_CAL	DC divider factor, 3.000 nominal · retentive
DF30	DISPLAY	What the T8C shows
DF31	REL_OFFSET	Captured zero
DF32 / 33	MIN_VAL · MAX_VAL	Minimum and maximum since reset
DF40 / 41	LIVE · LIVE_REL	Selected reading, then with REL applied
MODE (INTEGER)		
DS1	MODE	1 = AC V · 2 = DC V · 3 = Ω
DS2	MODE_LAST	Detects mode changes
BITS FROM THE T8C (WRITTEN BY THE PANEL)		
C1 / C2 / C3	HMI_HOLD · HMI_REL · HMI_MMRESET	Momentary buttons
C5	BEEP_EN	Toggle button, on at power-up
C6 / C7 / C8	HMI_ACV · HMI_DCV · HMI_OHM	Momentary mode buttons
STATUS BITS (READ BY THE PANEL)		
C10 / C11 / C12	ACV_OL · DCV_OL · OHM_OPEN	Over-range / open leads
C13 / C14	EXCITE_FLT · LOOP_FLT	Excitation under 5 V / VACT loop under 3.6 mA
C20 / C21	HOLD_ON · REL_ON	State latches
C30 – C36	HOLD_PULSE ... REL_RSTCMD	Toggle helper bits
C41	MODE_CHG	One-scan pulse on a mode change

ADDRESS	NICKNAME	WHAT IT IS
C44	SHOW_OL	The current mode is over-range or faulted
C51 / 52 / 53	MODE_ACV · MODE_DCV · MODE_OHM	Mode bits for rungs and screen visibility
T1	SAMPLE_100MS	Self-resetting 100 ms sample clock
SC1 / SC2	_Always_ON · _1st_SCAN	CLICK system bits

3 · Program

| normally open | / | normally closed | ↑ | rising edge [] compare contact

□ output instruction

INITIALIZE

1

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SC2
_1st_SCAN

[DF20 == 0.0]

Copy 3.0 → DF20 DCV_CAL

On the very first power-up, load the nominal 3:1 divider factor. After that DF20 keeps whatever you calibrate it to.

2

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SC2
_1st_SCAN

Set C5 BEEP_EN

Copy 2 → DS1 MODE

Start with the beeper enabled and in DC V mode, the safest mode to power up in.

3

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SC1
_Always_ON

Copy 10.0 → DF5 AO_EXCITE

Hold the ohms excitation output at 10 V.

SAMPLE CLOCK

4

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T1
SAMPLE_100MS

TMR T1 · preset 100 ms SAMPLE_100MS

Self-resetting timer: T1 is on for exactly one scan every 100 ms, and that pulse paces the averaging and MIN/MAX.

SCALE TO ENGINEERING UNITS

5

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SC1
_Always_ON

Math DF10 = (DF1 - 1.0) × 125.0 ACV_RAW

VACT: 1 V (4 mA) = 0 VAC and 5 V (20 mA) = 500 VAC.

6

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SC1
_Always_ON

Math DF11 = DF2 × DF20 DCV_RAW

DC volts = terminal volts × divider ratio (3.000 nominal).

7

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SC1
_Always_ON

Math DF16 = DF4 × 0.99 OHM_OPEN_THR

Anything above 99% of the actual excitation counts as open leads (about 200 kΩ).

8

[DF3 < DF16]

[DF4 > 5.0]

Math $DF12 = 2000.0 \times DF3 \div (DF4 - DF3)$ OHM_RAW

Ratio ohmmeter with the 2 kΩ reference. Only calculated when the leads aren't open and the excitation is healthy, so it never divides by zero.

OVER-RANGE & FAULT FLAGS

9

[DF3 >= DF16]

Out C12 OHM_OPEN

Ω: leads open or resistance above range.

10

[DF4 <= 5.0]

Out C13 EXCITE_FLT

Excitation collapsed. Look for a shorted probe on a live circuit, a tripped PTC2 or a blown F4.

11

[DF10 > 520.0]

Out C10 ACV_OL

AC V above the VACT's range. It caps at 24 mA, about 625 V indicated.

12

[DF1 < 0.9]

Out C14 LOOP_FLT

Loop current under 3.6 mA means the VACT is unpowered or the loop is broken or reversed.

13

[DF2 > 9.98]

Out C11 DCV_OL

DC input at the 30 V ceiling.

AVERAGING (EVERY 100 MS)

14


SAMPLE_100MS

Math $DF13 = DF13 + (DF10 - DF13) \times 0.3$ ACV_AVGMath $DF14 = DF14 + (DF11 - DF14) \times 0.3$ DCV_AVGMath $DF15 = DF15 + (DF12 - DF15) \times 0.3$ OHM_AVG

First-order filter. It settles about 90% in 0.6 s. Raise 0.3 for faster response, lower it for a steadier display.

MODE SELECT

15


HMI_ACV

Copy 1 → DS1 MODE

16


HMI_DCV

Copy 2 → DS1 MODE

17


HMI_OHM

Copy 3 → DS1 MODE

The three panel mode buttons.

18

[DS1 == 1]

Out C51 MODE_ACV

19

[DS1 == 2]

Out C52 MODE_DCV



One bit per mode, for the rungs below and for screen visibility and lamps on the T8C.

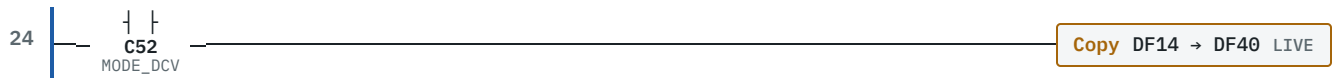
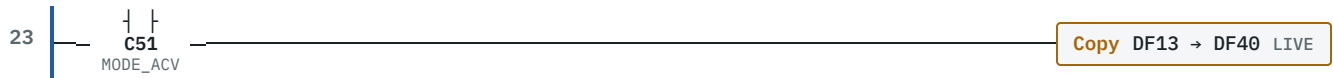


One-scan pulse whenever the mode changes.



Changing modes cancels HOLD and REL, so a stale zero or held reading never carries into another function.

SELECT LIVE READING



Route the active mode's filtered value to LIVE.



Show "OL" on the panel when the active function is over-range or faulted.

HOLD TOGGLE (TOUCH BUTTON)



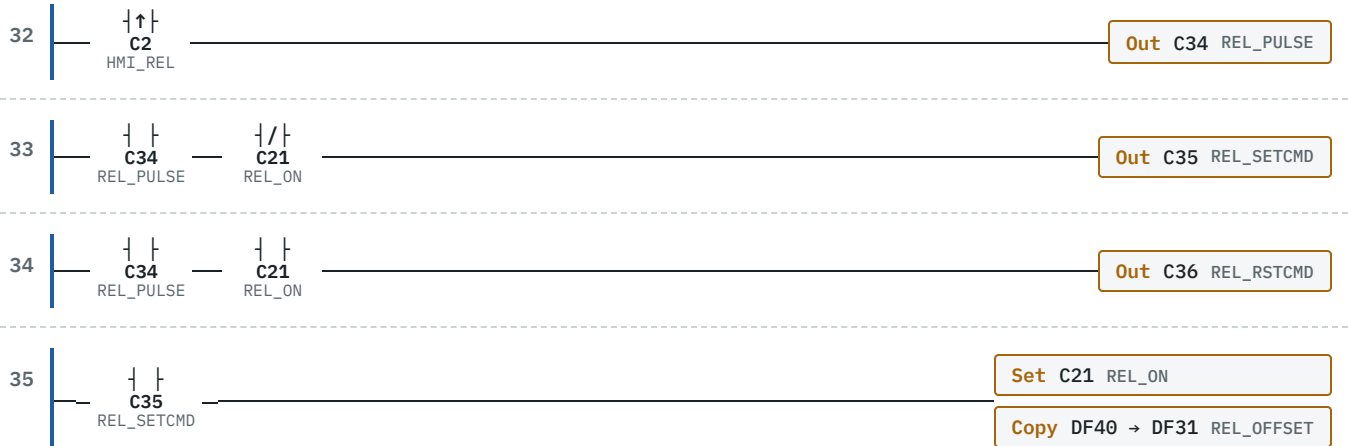
The HOLD touch button gives a one-scan pulse.



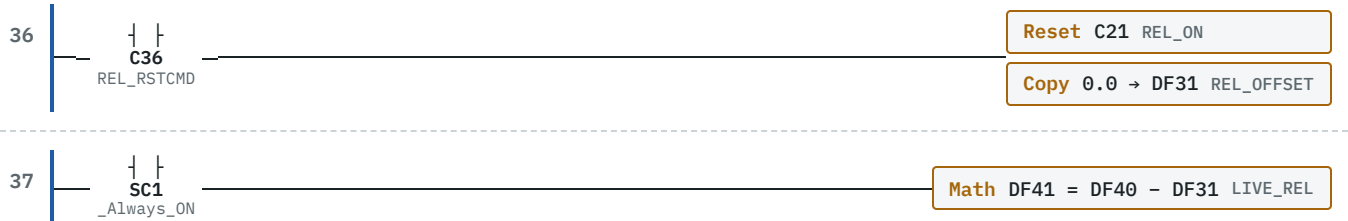
Decide the direction before changing the latch, so it flips exactly once per press.



REL (ZERO) TOGGLE (TOUCH BUTTON)



Turning REL on captures the present reading as the zero. With the leads shorted in Ω , that nulls out the leads and PTC2.



DISPLAY

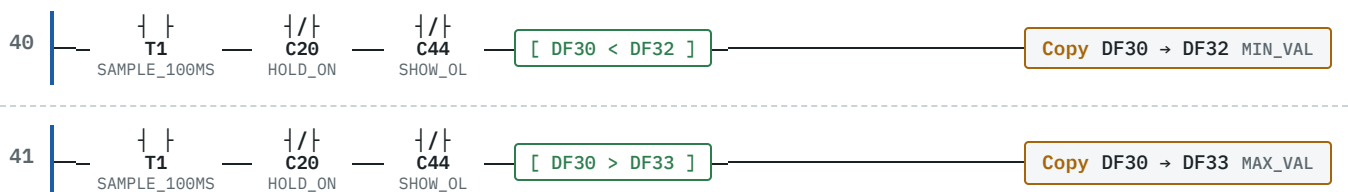


While HOLD is on this rung stops copying, so DF30 freezes at the last value.

MIN / MAX

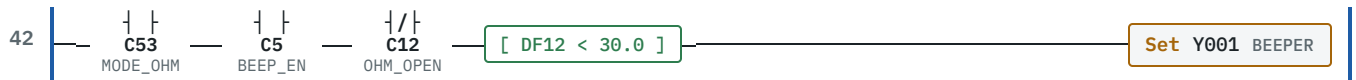


Reset both to the present reading on the button, a mode change, or power-up.



Track every 100 ms, but not while held or over-range, so an OL spike doesn't get recorded as a MAX.

CONTINUITY BEEPER



Uses the unfiltered reading so the beep is instant.



Turns off above 35 Ω . The 5 Ω gap between on and off keeps the buzzer from chattering on a borderline connection.

Order matters. Enter the rungs in this order. The toggles, mode-change pulse and REL capture depend on the earlier rungs running first in the same scan. CLICK adds its own END rung after the last one.

4 · C-more EA7-T8C tags

Panel setup

- **PLC protocol:** AutomationDirect CLICK, serial, 38400 baud, odd parity, 8 data bits, 1 stop bit, node 1. Set CLICK Port 2 (Setup → Com Port Setup) to Modbus RTU with the same settings.
- **Tags:** create them with the CLICK addresses exactly as shown. Use data type *Floating PT 32* for DF, *Signed Int 16* for DS, and *Discrete* for C.
- **Reading display:** make three numeric displays tied to DF30, stacked in the same spot. Give AC V 1 decimal, DC V 3 decimals, and Ω 1 decimal, and set each one's visibility from C51, C52 or C53.
- **OL overlay:** put a red "OL" text object on top, visible when C44 is on.
- **Trend graph:** plot DF30, sampling every 0.5 s over 60 s.

Objects → addresses

DF30	Main reading
DF32 / DF33	MIN / MAX displays
C6 · C7 · C8	Mode buttons (momentary)
C51 · C52 · C53	Mode button lamps and units text
C1 → C20	HOLD button · lamp
C2 → C21	REL button · lamp
C3	MIN/MAX reset (momentary)
C5	Beep on/off (toggle)
C13 · C14	"Excitation fault" · "Check VACT loop" alarms
DF20	DC calibration entry on a hidden setup screen

5 · First run

1. Program through Port 1. The panel stays connected on Port 2 the whole time. Put the CPU in RUN.
2. Open a Data View on DF1–DF5. You should see DF4 $\approx$ 10.0 and DF1 $\approx$ 1.00 (4 mA, 0 VAC), and C14 off.
3. DC mode: put a known 24 V on the DC jacks. If DF30 is off from your reference meter, set DF20 = $24.00 \div \text{DF2}$.
4. Ω mode, leads shorted: press REL. It should read 0.0 and the beeper sounds. Put a 1 k Ω resistor on the leads and check it reads about 1000.
5. AC mode: check against your handheld on 120 V first, then 480 V.